

Qualification Pack



Fortified Rice Technician

QP Code: FIC/Q1010

Version: 2.0

NSQF Level: 4

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Qualification Pack

Contents

FIC/Q1010: Fortified Rice Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
FIC/N9026: Prepare for production	5
FIC/N1036: Carry out production of fortified rice	10
FIC/N9906: Apply food safety guidelines in Food Processing	20
DGT/VSQ/N0101: Employability Skills (30 Hours)	27
Assessment Guidelines and Weightage	32
<i>Assessment Guidelines</i>	32
<i>Assessment Weightage</i>	33
Acronyms	34
Glossary	35

Qualification Pack

FIC/Q1010: Fortified Rice Technician

Brief Job Description

A fortified rice technician is responsible for producing fortified rice through extrusion method by operating extruder in synchronization with various tools and equipment as per the standard operating procedures of the organization to achieve uniform quality product with consistent output.

Personal Attributes

A Fortified Rice Technician must have the ability to plan, organize, prioritize the work. The individual must possess reading, writing, communication and problem-solving skills. In addition, the individual must be able to apply basic mathematical calculations, handle pressure, practice personal and professional hygiene.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [FIC/N9026: Prepare for production](#)
2. [FIC/N1036: Carry out production of fortified rice](#)
3. [FIC/N9906: Apply food safety guidelines in Food Processing](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Food Processing
Sub-Sector	Food Grain Milling
Occupation	Processing-Food Grain Milling (including oilseeds)
Country	India
NSQF Level	4
Credits	13
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL

Qualification Pack

Minimum Educational Qualification & Experience	12th grade Pass (or Equivalent) OR 10th grade pass with 3 Years of experience in the Food Industry OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience in the Food Industry OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the Food Industry
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	2.0
Reference code on NQR	QG-04-FI-04001-2025-V2-FICSI
NQR Version	2.0

Qualification Pack

FIC/N9026: Prepare for production

Description

This NOS unit is about performing various tasks prior to production in the food processing industry.

Scope

The scope covers the following :

- Plan for production
- Clean and maintain work area, machineries, and tools for production
- Organize for production

Elements and Performance Criteria

Plan for production

To be competent, the user/individual on the job must be able to:

- PC1.** identify work requirements by obtaining instructions from the supervisor. Instructions: process chart, product flow chart, formulation, chart, etc.
- PC2.** plan and prioritize tasks as per work schedule. Tasks: inspect, clean, maintain, verify, etc.
- PC3.** estimate manpower and material requirements as per work requirement. Material: raw materials and packaging materials
- PC4.** ensure required quantity of raw materials, packaging materials, equipment, and manpower for production
- PC5.** plan capacity utilization of machinery with respect to the processing time, production order, and batch size for each product

Clean and maintain work area, machineries, and tools for production

To be competent, the user/individual on the job must be able to:

- PC6.** clean and maintain the work area as per organizational procedures
- PC7.** clean and maintain the machines and tools and sanitize them as per the organization's specifications and standards
- PC8.** dispose of the waste material at designated place safely. Waste material: hazardous waste, food waste, packaging waste, etc.
- PC9.** inspect the tools, equipment, and machinery to ascertain suitability for use
- PC10.** report information such as faulty tools and equipment to the concerned authority

Organize for production

To be competent, the user/individual on the job must be able to:

- PC11.** organize tools and equipment
- PC12.** receive and organize production materials appropriately. Production materials: raw materials, packaging materials, etc.
- PC13.** allot responsibilities/work to the assistants and helpers

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** production planning process
- KU2.** analysis and interpretation of various process charts, product flow charts, etc.
- KU3.** resource management process
- KU4.** procedure to estimate manpower and raw material
- KU5.** capacity utilization calculation
- KU6.** organizational policies and SOP on cleanliness
- KU7.** operating procedure and general maintenance of food production machineries
- KU8.** waste management procedures
- KU9.** methods to inspect tools, equipment and machinery
- KU10.** procedure to allot work or responsibility to the team

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret organizational policies, SOP, production charts, etc.
- GS2.** communicate effectively with subordinates as well as supervisors
- GS3.** plan and prioritize various tasks
- GS4.** be always punctual and courteous
- GS5.** organize all process/equipment manuals to access information easily
- GS6.** discuss task lists, schedules, and activities with the senior/supervisor

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan for production</i>	11	25	-	-
PC1. identify work requirements by obtaining instructions from the supervisor. Instructions: process chart, product flow chart, formulation, chart, etc.	3	6	-	-
PC2. plan and prioritize tasks as per work schedule.Tasks: inspect, clean, maintain, verify, etc.	2	5	-	-
PC3. estimate manpower and material requirements as per work requirement. Material: raw materials and packaging materials	2	4	-	-
PC4. ensure required quantity of raw materials, packaging materials, equipment, and manpower for production	2	5	-	-
PC5. plan capacity utilization of machinery with respect to the processing time, production order, and batch size for each product	2	5	-	-
<i>Clean and maintain work area, machineries, and tools for production</i>	14	32	-	-
PC6. clean and maintain the work area as per organizational procedures	3	7	-	-
PC7. clean and maintain the machines and tools and sanitize them as per the organization's specifications and standards	3	7	-	-
PC8. dispose of the waste material at designated place safely. Waste material: hazardous waste, food waste, packaging waste, etc.	3	7	-	-
PC9. inspect the tools, equipment, and machinery to ascertain suitability for use	3	6	-	-
PC10. report information such as faulty tools and equipment to the concerned authority	2	5	-	-
<i>Organize for production</i>	5	13	-	-
PC11. organize tools and equipment	2	7	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. receive and organize production materials appropriately. Production materials: raw materials, packaging materials, etc.	2	4	-	-
PC13. allot responsibilities/work to the assistants and helpers	1	2	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	FIC/N9026
NOS Name	Prepare for production
Sector	Food Processing
Sub-Sector	Generic
Occupation	Production
NSQF Level	3
Credits	TBD
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

FIC/N1036: Carry out production of fortified rice

Description

This unit is about producing fortified rice kernels by the method of extrusion and then blending it by using different machineries to produce fortified rice as per the specifications and standards of the organization.

Scope

The scope covers the following :

- Prepare for selection of raw materials
- Perform production of fortified rice kernels
- Perform production of blended fortified rice
- Monitor losses during extrusion
- Ensure proper storage and packaging of fortified rice

Elements and Performance Criteria

Prepare for selection of raw materials

To be competent, the user/individual on the job must be able to:

- PC1.**
 - identify food grade vitamin and mineral premix
 - premix: Vitamin B12, Iron, folic acid
- PC2.** procure broken rice after rice milling to prepare for fortification process by checking the specification limit
- PC3.**
 - ensure the obtained rice meets the parameters as per standards of normal rice
 - parameters: strong to sustain shelf life, cooking quality, etc.
- PC4.** pass the broken rice through a vibro separator to check the presence of foreign material as well as through magnets to assure absence of any magnetic particles
- PC5.** discard the raw material which does not meet the standards
- PC6.** ensure the obtained premix meets the standards set by FSSAI
- PC7.**
 - arrange and store the ingredients properly in a designated area
 - ingredients: Rice, food grade vitamin and mineral premix, FSSAI approved acid regulators and emulsifiers, potable water
- PC8.** follow the flowchart prepared by the organization to carry out the production process

Perform production of fortified rice kernels

To be competent, the user/individual on the job must be able to:

- PC9.** transfer the approved broken rice into a pulverizer to grind it further to a desired size
- PC10.**
 - sieve the broken rice to check the standard particle size and removal of foreign particles
 - standard particle size: 150 µm
- PC11.** ensure the moisture content of broken rice before the mixing process should be in a range 11-12%
- PC12.** mix the vitamin and mineral premix with the broken rice in a flour mixer layer by layer
- PC13.** transfer the mixture into a preconditioner to add steam into the mixture for partial gelatinization of starch

Qualification Pack

- PC14.** transfer the mixture into a feeding hopper with a screw conveyor or manually to incorporate water in it
- PC15.** monitor the temperature and moisture content of the mixture in the hopper with a temperature moisture detector gun
- PC16.** transfer the moist fortified flour into a twin screw extruder for partial cooking and extruding rice shaped kernels
- PC17.** ensure the main, feeding and cutting motor works properly in maintaining the shape and size of the kernels
- PC18.** • monitor the extrusion parameters to achieve consistency throughout the process
• parameters: Feed rate, temperature and rotation speed of motor
- PC19.** measure the length, thickness and width of the extruded kernel to match the specification
- PC20.** cool the extruded kernels to remove moisture and temperature in a vibratory pan conveyor
- PC21.** transfer the extruded fortified kernels into five layered dryer through a pneumatic line or manually to reduce the moisture content of fortified rice kernels to 11-12%
- PC22.** maintain the temperature parameter of the vibratory dryer
- PC23.** transfer the fortified rice kernels from the dryer into a vibro separator to check and remove any clumps and extraneous matter
- PC24.** pass the kernels through a permanent bar magnet to check the presence of metal in fortified rice kernels
- PC25.** send the sample to quality lab for analysis analysis: moisture, nutritional content, (vitamin and mineral in premix) kernel size, colour etc.

Perform production of blended fortified rice

To be competent, the user/individual on the job must be able to:

- PC26.** fill one hopper with fortified rice kernel and other with normal rice for batch blending
- PC27.** ensure the dosing system feeds the fortified rice kernel to normal rice in a designated ratio or as required to meet the fortified rice specification
- PC28.** set the combination of dosing parameters of a vibratory feeder in a continuous blending process
- PC29.** ensure the helical ribbon blender blends the mixture to produce homogenous fortified rice
- PC30.** ensure the steps of blending are performed just prior to packaging the fortified rice
- PC31.** • send the sample of finished product to quality lab to check quality parameters
• parameters: moisture content, texture, broken kernels, organoleptic quality, pesticide residue, etc.

Monitor losses during extrusion

To be competent, the user/individual on the job must be able to:

- PC32.** • monitor the process loss by controlling the factors affecting stability during the fortification process
• process loss: applied heat, humidity during heating, drying steps, presence, or absence of air
- PC33.** monitor the stability of fortified rice during storage by using antioxidant for stabilization of vitamin A
- PC34.** ensure the mixture is properly blended during mixing to avoid washing loss
- PC35.** monitor the loss of water-soluble vitamins during cooking process by using FSSAI approved emulsifiers

Ensure proper storage and packaging of fortified rice

Qualification Pack

To be competent, the user/individual on the job must be able to:

- PC36.** dispose the rejected batch safely following the standard practice of the organization
- PC37.** pack the fortified rice in two-layer bag with inner layer of poly lining or equivalent packaging material
- PC38.** ensure the packed bag passes the butt drop and flat drop test
- PC39.** perform shelf-life study of fortified rice to maintain its shelf stability
- PC40.** report any non-conformity in the process to the higher authority
- PC41.**
 - ensure the mandatory information are printed on the label of the bag
 - Information: Name, Net weight, name of supplier, production date, list of ingredients in descending order, country of origin, lot identification, storage instructions, '+F' logo, etc.
- PC42.** store the fortified rice under dry, ventilated, and hygienic conditions for further transportation
- PC43.** store the tools, equipment and materials used for production safely in designated places
- PC44.** maintain the tools, equipment and materials used for production as per standard practices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and procedures related to fortified rice production
- KU2.** types of vitamin and mineral premix
- KU3.** tasks to be performed and processes involved in fortification production
- KU4.** types of tools, equipment and machinery used in fortified rice production
- KU5.** method to measure parameters like antioxidants, emulsifiers, premix, etc.
- KU6.** various signs of damaged kernels and their inspection techniques
- KU7.** technique for performing hot or cold extrusion
- KU8.** process of blending and different types of blending
- KU9.** importance of preconditioning of kernels
- KU10.** different types of fortification like dusting and coating
- KU11.** instruments used for monitoring temperature and moisture content of rice
- KU12.** process of checking the dosing system
- KU13.** types of dryer and their parameters
- KU14.** optimal conditions required for drying the fortified kernels
- KU15.** different tests performed to check the quality
- KU16.** different losses during production and how to overcome them
- KU17.** procedure for storing, packing and labelling the fortified rice and the materials used in the process
- KU18.** instances that need to be reported to the concerned personnel and the methods used
- KU19.** types of information that is documented and the organisational practices for documenting them
- KU20.** Food Safety and Standards Authority of India (FSSAI) guidelines for production, packaging, labelling, storing, handling fortified rice and raw material

Qualification Pack

- KU21.** procedure to store the tools, equipment and materials used for production safely
- KU22.** procedure to maintain the tools, equipment and materials used for fortified rice production
- KU23.** how to coordinate with vendors for vitamin and mineral premix transportation and delivery
- KU24.** which matter to be reported to the higher authority and when
- KU25.** safe disposal of waste and unwanted materials at the workplace
- KU26.** SOP to record the information (such as type of FRK produced, its characteristics, batches produced, quantity to be dispatched, etc.) as required in the job

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note the raw materials used for production and the finished products produced
- GS2.** note the readings of the process parameters and necessary information to fill the process chart
- GS3.** note down observations (if any) related to the process
- GS4.** write information documents to internal departments/ internal teams
- GS5.** note down the data for ERP or as required by the organization
- GS6.** read and interpret the process required for producing various types of products
- GS7.** read and interpret and process flowcharts for all products produced
- GS8.** read equipment manuals and process documents to understand the equipment operations
- GS9.** read internal information documents sent by internal teams
- GS10.** discuss task lists, schedules, and activities with the higher authority
- GS11.** effectively communicate with the team members
- GS12.** question the higher authority to understand the nature of the problem and to clarify
- GS13.** note down the information gathered from the higher authority
- GS14.** attentively listen and comprehend the information given by the speaker
- GS15.** communicate clearly with the internal team and cross-department team on the issues
- GS16.** plan and organize the work order and jobs received from the higher authority
- GS17.** organize raw materials and packaging materials required for making fortified rice
- GS18.** plan and prioritize work based on the instructions received from the higher authority
- GS19.** plan to utilize time and equipment effectively
- GS20.** organize all process / equipment manuals to access information easily
- GS21.** support higher authority in solving problems by detailing out problems
- GS22.** discuss the possible solutions with the higher authority for problem-solving
- GS23.** apply information about maintenance processes & knowledge about tools
- GS24.** use common sense and make judgments on a day-to-day basis
- GS25.** use reasoning skills to identify and resolve basic problems
- GS26.** use intuition to detect any potential problems which could arise during operations
- GS27.** use acquired knowledge of the process for identifying and handling issues
- GS28.** Identify critical points in daily tasks and apply control measures to solve the issue

Qualification Pack

GS29. support the higher authority in scheduling tasks for helper(s)

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for selection of raw materials</i>	5	10	-	5
PC1. • identify food grade vitamin and mineral premix • premix: Vitamin B12, Iron, folic acid	-	-	-	-
PC2. procure broken rice after rice milling to prepare for fortification process by checking the specification limit	-	-	-	-
PC3. • ensure the obtained rice meets the parameters as per standards of normal rice • parameters: strong to sustain shelf life, cooking quality, etc.	-	-	-	-
PC4. pass the broken rice through a vibro separator to check the presence of foreign material as well as through magnets to assure absence of any magnetic particles	-	-	-	-
PC5. discard the raw material which does not meet the standards	-	-	-	-
PC6. ensure the obtained premix meets the standards set by FSSAI	-	-	-	-
PC7. • arrange and store the ingredients properly in a designated area • ingredients: Rice, food grade vitamin and mineral premix, FSSAI approved acid regulators and emulsifiers, potable water	-	-	-	-
PC8. follow the flowchart prepared by the organization to carry out the production process	-	-	-	-
<i>Perform production of fortified rice kernels</i>	10	20	-	5
PC9. transfer the approved broken rice into a pulverizer to grind it further to a desired size	-	-	-	-
PC10. • sieve the broken rice to check the standard particle size and removal of foreign particles • standard particle size: 150 µm	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure the moisture content of broken rice before the mixing process should be in a range 11-12%	-	-	-	-
PC12. mix the vitamin and mineral premix with the broken rice in a flour mixer layer by layer	-	-	-	-
PC13. transfer the mixture into a preconditioner to add steam into the mixture for partial gelatinization of starch	-	-	-	-
PC14. transfer the mixture into a feeding hopper with a screw conveyor or manually to incorporate water in it	-	-	-	-
PC15. monitor the temperature and moisture content of the mixture in the hopper with a temperature moisture detector gun	-	-	-	-
PC16. transfer the moist fortified flour into a twin screw extruder for partial cooking and extruding rice shaped kernels	-	-	-	-
PC17. ensure the main, feeding and cutting motor works properly in maintaining the shape and size of the kernels	-	-	-	-
PC18. - monitor the extrusion parameters to achieve consistency throughout the process - parameters: Feed rate, temperature and rotation speed of motor	-	-	-	-
PC19. measure the length, thickness and width of the extruded kernel to match the specification	-	-	-	-
PC20. cool the extruded kernels to remove moisture and temperature in a vibratory pan conveyor	-	-	-	-
PC21. transfer the extruded fortified kernels into five layered dryer through a pneumatic line or manually to reduce the moisture content of fortified rice kernels to 11-12%	-	-	-	-
PC22. maintain the temperature parameter of the vibratory dryer	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. transfer the fortified rice kernels from the dryer into a vibro separator to check and remove any clumps and extraneous matter	-	-	-	-
PC24. pass the kernels through a permanent bar magnet to check the presence of metal in fortified rice kernels	-	-	-	-
PC25. send the sample to quality lab for analysis analysis: moisture, nutritional content, (vitamin and mineral in premix) kernel size, colour etc.	-	-	-	-
<i>Perform production of blended fortified rice</i>	5	5	-	-
PC26. fill one hopper with fortified rice kernel and other with normal rice for batch blending	-	-	-	-
PC27. ensure the dosing system feeds the fortified rice kernel to normal rice in a designated ratio or as required to meet the fortified rice specification	-	-	-	-
PC28. set the combination of dosing parameters of a vibratory feeder in a continuous blending process	-	-	-	-
PC29. ensure the helical ribbon blender blends the mixture to produce homogenous fortified rice	-	-	-	-
PC30. ensure the steps of blending are performed just prior to packaging the fortified rice	-	-	-	-
PC31. • send the sample of finished product to quality lab to check quality parameters • parameters: moisture content, texture, broken kernels, organoleptic quality, pesticide residue, etc.	-	-	-	-
<i>Monitor losses during extrusion</i>	5	5	-	5
PC32. • monitor the process loss by controlling the factors affecting stability during the fortification process • process loss: applied heat, humidity during heating, drying steps, presence, or absence of air	-	-	-	-
PC33. monitor the stability of fortified rice during storage by using antioxidant for stabilization of vitamin A	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. ensure the mixture is properly blended during mixing to avoid washing loss	-	-	-	-
PC35. monitor the loss of water-soluble vitamins during cooking process by using FSSAI approved emulsifiers	-	-	-	-
<i>Ensure proper storage and packaging of fortified rice</i>	5	10	-	5
PC36. dispose the rejected batch safely following the standard practice of the organization	-	-	-	-
PC37. pack the fortified rice in two-layer bag with inner layer of poly lining or equivalent packaging material	-	-	-	-
PC38. ensure the packed bag passes the butt drop and flat drop test	-	-	-	-
PC39. perform shelf-life study of fortified rice to maintain its shelf stability	-	-	-	-
PC40. report any non-conformity in the process to the higher authority	-	-	-	-
PC41. - ensure the mandatory information are printed on the label of the bag - Information: Name, Net weight, name of supplier, production date, list of ingredients in descending order, country of origin, lot identification, storage instructions, '+F' logo, etc.	-	-	-	-
PC42. store the fortified rice under dry, ventilated, and hygienic conditions for further transportation	-	-	-	-
PC43. store the tools, equipment and materials used for production safely in designated places	-	-	-	-
PC44. maintain the tools, equipment and materials used for production as per standard practices	-	-	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N1036
NOS Name	Carry out production of fortified rice
Sector	Food Processing
Sub-Sector	Food Grain Milling
Occupation	Processing-Food Grain Milling (including oilseeds)
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

FIC/N9906: Apply food safety guidelines in Food Processing

Description

This unit covers the essential components of food safety, Good Manufacturing Practices (GMP), and personal hygiene in the food industry. It emphasizes the importance of individuals working in the food industry in protecting the health and well-being of consumers by following food safety protocols and procedures and ensuring the production of safe and high-quality food products.

Scope

The scope covers the following :

- Apply personal hygiene and follow Good Manufacturing practices at the workplace.
- Implement Food Safety and pre-requisite programs (PRP) at the workplace.

Elements and Performance Criteria

Apply personal hygiene and follow Good Manufacturing practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** PC1. follow a site relevant documented procedure for Personal Hygiene and Visitor/ Contractor rules.
- PC2.** PC2. follow work instructions at levels of employees inside a food manufacturing site and ensure that the relevant instructions are well communicated and being followed at the fixed timelines.
- PC3.** PC3. ensure timely participate and carry out the relevant training and awareness sessions on personal hygiene, GMP, and related topics.
- PC4.** PC4. ensure timely medical examination from a prescribed and authorized doctor and comply with the guidelines of Schedule IV as described in Food Safety Standard Authority of India (FSSAI) guidelines.
- PC5.** PC5. fill in data in the daily monitoring checklist related to personal hygiene, food safety, and GMP.
- PC6.**
 - PC6. follow a site-relevant documented procedure and area-wise work instructions for Good Manufacturing Practices (GMP) to be followed on the site.
 - procedure: Hand washing requirements, Gowning & De gowning protocols, cleaning, and sanitation of employee lockers, follow the protocols as laid down in the different categories of processing areas like Low Risk, High Risk, High Care areas, etc.
- PC7.** PC7. follow all validated Do's & Don'ts inside a food manufacturing firm.
- PC8.** PC8. follow man and materials movement throughout the production facility, to restrict unwanted hazards to cross-contaminate the products which are being manufactured in the facility.
- PC9.** PC9. refer to the process flow charts, HACCP summary plan, and critical process parameters in each and respective areas of the production line.
- PC10.** PC10. identify the material requirements such as manufacturing equipments, Utensils, and other processing aids, cleaning chemicals, and cleaning work instructions in all the relevant areas of the manufacturing facility. Also, a special focus shall be given to Allergens and their risks. Wherever required, the allergen requirements shall be separately addressed.

Qualification Pack

- PC11.** PC11. ensure to properly tag and number all the equipment, machinery, tools, and other processing aids to keep proper traceability of the product being manufactured and handled at the site.
- PC12.** PC12. follow and implement all training and awareness guidelines in the manufacturing area and regularly participate in training effectiveness for evaluation.
- PC13.** PC13. participate in audits and address the aspects of Good Manufacturing Procedures, personal hygiene, and food safety.
- PC14.** PC14. ensure the record keeping and documentation such as Daily Monitoring Sheets, Batch Traceability Records, machine records, product parameters, process control parameters, etc.

Implement food safety practices at the workplace

To be competent, the user/individual on the job must be able to:

- PC15.** PC15. maintain updated facilities, equipment, and tool and design requirements to minimize the risks associated with the products being handled at the site.
- PC16.** PC16. follow the instruction in the raw and packaging materials warehouse and ensure receiving material parameters match all the laid requirements. parameters: Incoming vehicles Visual report, storage, and handling requirements, hazardous and non-hazardous goods, allergens, cross-contamination risks, Quarantine, Accepted & rejected goods, monitoring temperature and humidity, etc.
- PC17.** PC17. follow FSSAI Schedule IV requirements related to Pest Control, Cleaning, and Sanitation, Utilities, Waste Disposal, Prevention of Cross-Contamination, allergen management, corrective action, preventive actions, food operation control etc.
- PC18.** PC18. ensure timely check of the critical control points and product parameters.
- PC19.** PC19. record keeping and documentation such as daily monitoring sheets, cleaning sheets, parameters, etc.
- PC20.** PC20. report any food safety and GMP issue to the supervisor, if any.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of personal hygiene, GMP, visitors & contractor's rules. Associated risk in case of deviation from the standard policies and how the requirement is linked with the site's FSSAI License.
- KU2.** KU2. importance of training and work instruction delivered by the supervisors.
- KU3.** KU3. importance of filling the records and checklists, formats and how to ensure that the timely and effective completion is achieved.
- KU4.** KU4. knowledge of trainings and skills required to perform in food processing premises.
- KU5.** KU5. understand FSSAI Schedule IV requirements of food handlers and PRPs within the processing area
- KU6.** KU6. importance of timely medical examinations and awareness of communicable diseases
- KU7.** Understanding of Do's & Don'ts, intellect mindset to understand the visual illustrations
- KU8.** KU8. understanding about Site Zoning plans.
- KU9.** KU9. awareness of layout which would help to demarcate the defined movements of RM, PM, FG, and wastes generated during the processing of goods. This one lays a framework to launch Good Manufacturing Practices (GMP) successfully and effectively on site.

Qualification Pack

- KU10.** KU10. understand the manufacturing process, product parameters and process control parameters such as CCPs
- KU11.** KU11. understanding about Hazard Analysis and Critical Control Points (HACCP)
- KU12.** KU12. understanding about Allergens and their types and controls to monitor effective handling of allergen raw materials on site.
- KU13.** KU13. basic understanding of traceability and mock recall
- KU14.** KU14. awareness about Internal & external Audits
- KU15.** KU15. understanding for RCA CAPA, cleaning and sanitation
- KU16.** KU16. awareness about record keeping and data monitoring in various sheets as per organizational requirement

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** GS1. read and comprehend basic content to read labels, charts, signages, symbols and product manuals
- GS2.** GS2. communicate with coworkers appropriately to clarify instructions and other issues
- GS3.** GS3. plan and organize the work schedule, work area, tools, equipment, and materials for improved productivity
- GS4.** GS4. plan and prioritize tasks as per work requirements
- GS5.** GS5. always be punctual and courteous
- GS6.** GS6. good observations and intellect mindset

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply personal hygiene and follow Good Manufacturing practices at workplace</i>	22	44	-	6
PC1. PC1. follow a site relevant documented procedure for Personal Hygiene and Visitor/ Contractor rules.	2	4	-	-
PC2. PC2. follow work instructions at levels of employees inside a food manufacturing site and ensure that the relevant instructions are well communicated and being followed at the fixed timelines.	2	4	-	2
PC3. PC3. ensure timely participate and carry out the relevant training and awareness sessions on personal hygiene, GMP, and related topics.	2	4	-	-
PC4. PC4. ensure timely medical examination from a prescribed and authorized doctor and comply with the guidelines of Schedule IV as described in Food Safety Standard Authority of India (FSSAI) guidelines.	2	4	-	-
PC5. PC5. fill in data in the daily monitoring checklist related to personal hygiene, food safety, and GMP.	2	4	-	-
PC6. - PC6. follow a site-relevant documented procedure and area-wise work instructions for Good Manufacturing Practices (GMP) to be followed on the site. - procedure: Hand washing requirements, Gowning & De gowning protocols, cleaning, and sanitation of employee lockers, follow the protocols as laid down in the different categories of processing areas like Low Risk, High Risk, High Care areas, etc.	2	4	-	2
PC7. PC7. follow all validated Do's & Don'ts inside a food manufacturing firm.	1	2	-	1
PC8. PC8. follow man and materials movement throughout the production facility, to restrict unwanted hazards to cross-contaminate the products which are being manufactured in the facility.	2	4	-	-
PC9. PC9. refer to the process flow charts, HACCP summary plan, and critical process parameters in each and respective areas of the production line.	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. PC10. identify the material requirements such as manufacturing equipments, Utensils, and other processing aids, cleaning chemicals, and cleaning work instructions in all the relevant areas of the manufacturing facility. Also, a special focus shall be given to Allergens and their risks. Wherever required, the allergen requirements shall be separately addressed.	2	4	-	-
PC11. PC11. ensure to properly tag and number all the equipment, machinery, tools, and other processing aids to keep proper traceability of the product being manufactured and handled at the site.	1	2	-	-
PC12. PC12. follow and implement all training and awareness guidelines in the manufacturing area and regularly participate in training effectiveness for evaluation.	1	2	-	-
PC13. PC13. participate in audits and address the aspects of Good Manufacturing Procedures, personal hygiene, and food safety.	1	2	-	-
PC14. PC14. ensure the record keeping and documentation such as Daily Monitoring Sheets, Batch Traceability Records, machine records, product parameters, process control parameters, etc.	1	2	-	-
<i>Implement food safety practices at the workplace</i>	8	16	-	4
PC15. PC15. maintain updated facilities, equipment, and tool and design requirements to minimize the risks associated with the products being handled at the site.	2	4	-	-
PC16. PC16. follow the instruction in the raw and packaging materials warehouse and ensure receiving material parameters match all the laid requirements. parameters: Incoming vehicles Visual report, storage, and handling requirements, hazardous and non-hazardous goods, allergens, cross-contamination risks, Quarantine, Accepted & rejected goods, monitoring temperature and humidity, etc.	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. PC17. follow FSSAI Schedule IV requirements related to Pest Control, Cleaning, and Sanitation, Utilities, Waste Disposal, Prevention of Cross-Contamination, allergen management, corrective action, preventive actions, food operation control etc.	2	4	-	2
PC18. PC18. ensure timely check of the critical control points and product parameters.	1	2	-	-
PC19. PC19. record keeping and documentation such as daily monitoring sheets, cleaning sheets, parameters, etc.	1	2	-	1
PC20. PC20. report any food safety and GMP issue to the supervisor, if any.	1	2	-	-
NOS Total	30	60	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	FIC/N9906
NOS Name	Apply food safety guidelines in Food Processing
Sector	Food Processing
Sub-Sector	Generic
Occupation	Generic
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
FIC/N9026.Prepare for production	30	70	-	-	100	30
FIC/N1036.Carry out production of fortified rice	30	50	-	20	100	45
FIC/N9906.Apply food safety guidelines in Food Processing	30	60	-	10	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	110	210	-	30	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualification Pack
TVET	Technical and Vocational Education and Training
MLF	Malolactic Fermentation
TA	Titrateable Acid
pH	Power of Hydrogen
FSSAI	Food Safety and Standards Authority of India
GMPs	Good Manufacturing Practices
HACCP	Hazard Analysis Critical Control Point
VACCP	Vulnerability Assessment Critical Control Points
TACCP	Threat Assessment Critical Control Points
RCA	Root Cause Analysis
CAPA	Corrective Action Preventive Action
PPE	Personal Protective Equipment
CPR	Cardio-Pulmonary Resuscitation
GHP	Good Hygiene Practices
SOPs	Standard Operating Procedures
PwD	People with Disabilities

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-Sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
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Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'.
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge & Understanding (KU)	Knowledge and Understanding (KU) are statements that together specify, the technical, generic, professional, and organizational specific knowledge that an individual needs in order to perform to the required standard.
Organizational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers has of their relevant areas of responsibility.
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